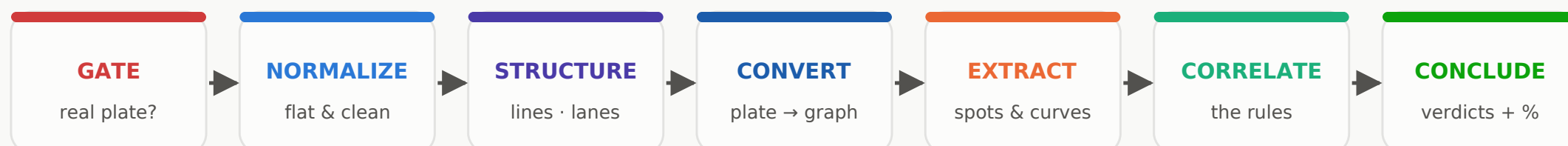


The TLC plate reading system

One photo in → chemist-grade answers out. Every stage shown working on our own plates.

Prepared by Tejas Ghatule



THREE PROMISES

Every number carries a confidence score with named reasons. The same image gives the same answer on every run (verified: two full runs, hash-identical). Every claim is replayable — when a result looks wrong, the trace shows the exact step where it went off-road. Hard rejection happens only twice: not a TLC plate, or not workable — decided upfront, with the reason.

WHAT MAKES THIS PAGE-PER-STAGE DOCUMENT DIFFERENT

Nothing here is a mock-up. Every input picture is one of our lab's real photos, and every output picture was produced by the prototype pipeline running on it. What you see is what the system does.

Real plate? Workable? — decided at the door

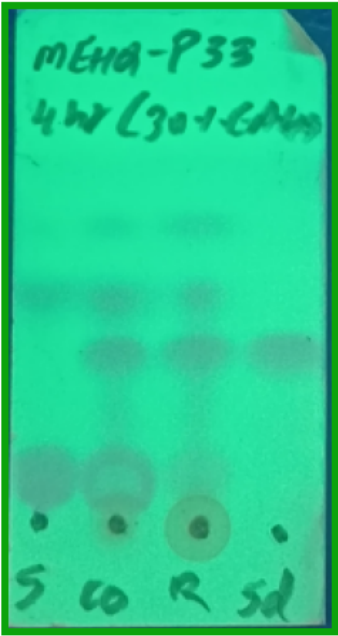
Only two hard stops in the whole system; everything else degrades with lowered confidence

screening plate



PASS — analysed

messy archive plate



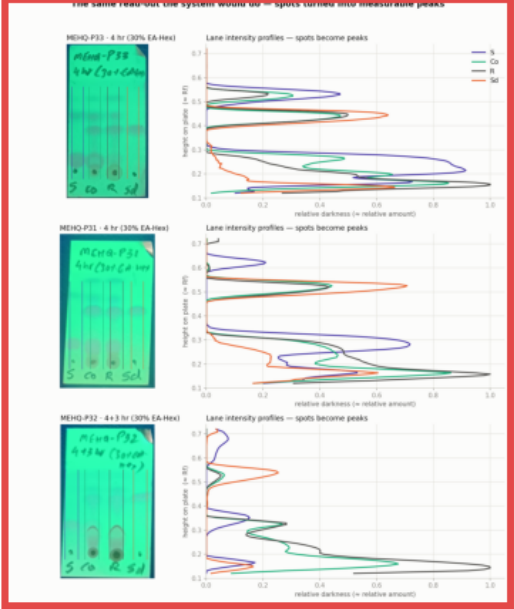
PASS — analysed
(penalties in scores)

not a plate



REJECT —
not UV-green

a chart image



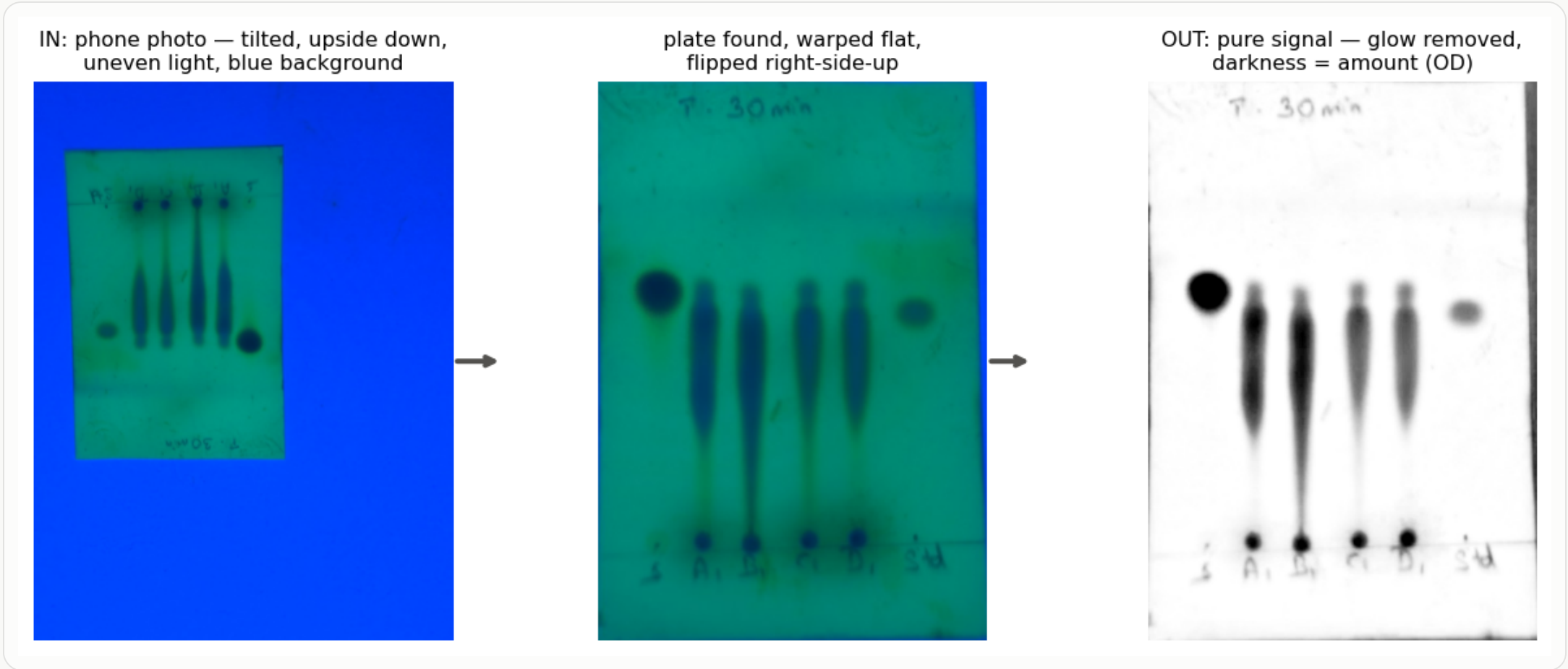
REJECT —
not UV-green, textured

IN: any image. OUT: pass (analysed — even messy plates), or an upfront rejection naming the reason. A genuine plate is never refused for being ugly; its penalties live in the confidence scores.

Checks (all computed from the image): UV-green background · low texture · lane-like structure · enough pixels · focus.

Any photo becomes a clean, flat plate

Find the plate, fix perspective, flip if upside down, remove the glow

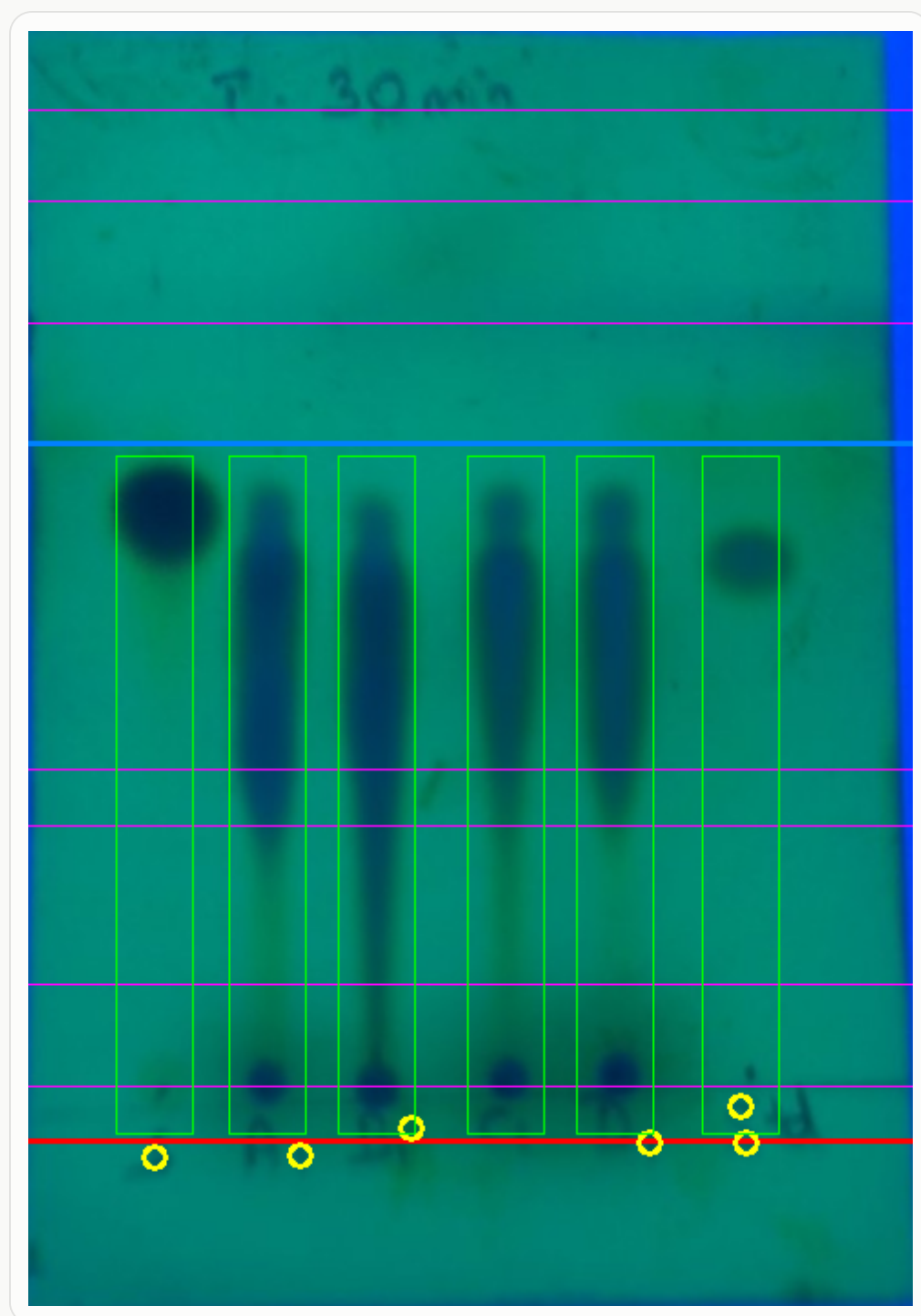


IN: a tilted, upside-down phone photo on a blue UV-box background (a real one of ours). OUT: the plate alone — straightened, right-side-up, illumination removed, so darkness now means amount and nothing else.

Tools: OpenCV segmentation + perspective warp · orientation from text/dot signals · iterative background fit → OD signal.

Lines, dots, lanes — the plate's coordinate system

Lane count is measured, never assumed; drawn lines found, virtual origin fitted when missing

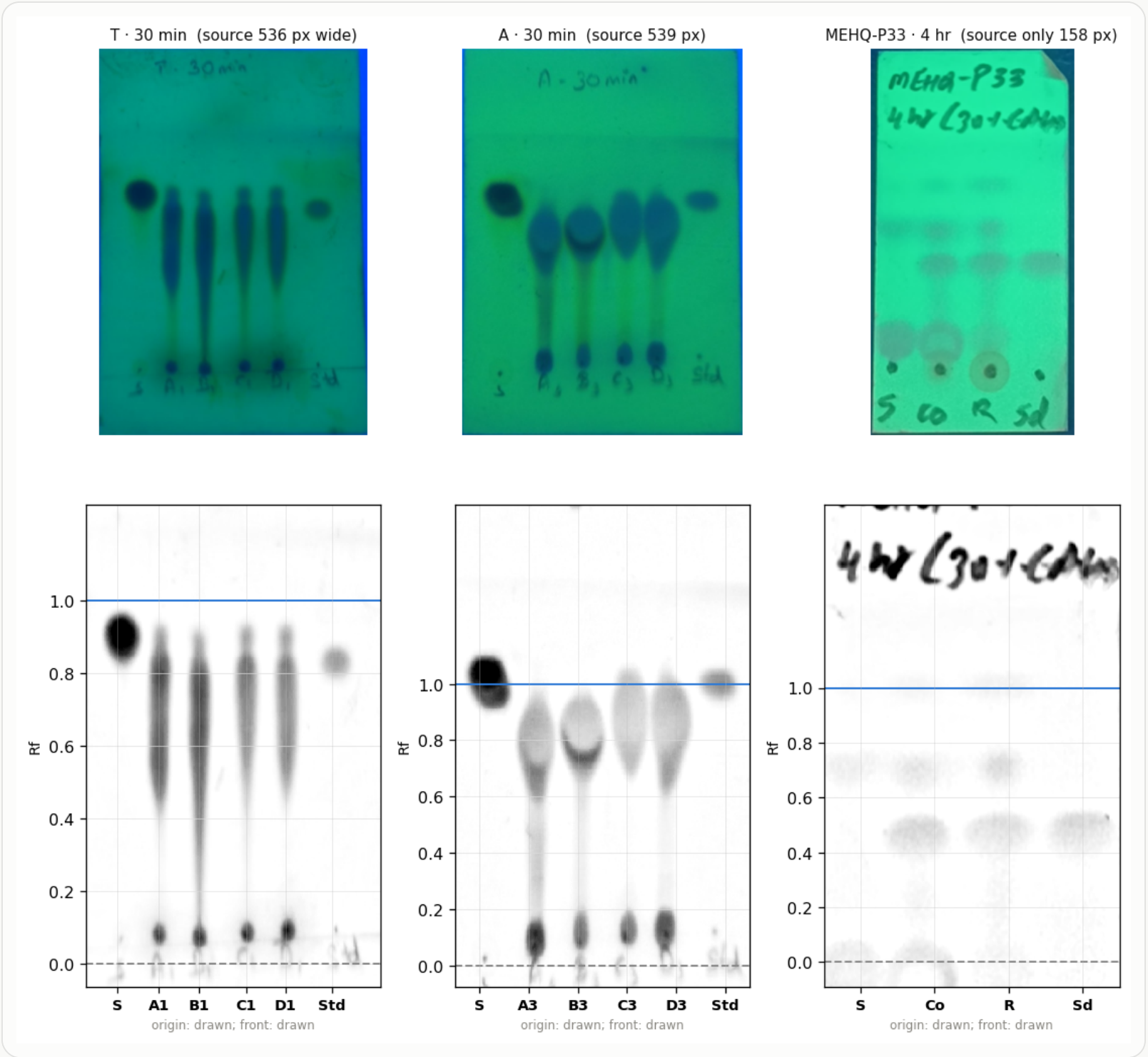


IN: the normalized plate. OUT: origin line (red — here the drawn pencil line was found through the spotting dots), solvent-front candidate (blue), origin dots (yellow), and 6 detected lanes (green) — the count came from the image.

Roles then resolve by lexicon → behaviour → one-tap user question (saved forever). 'S' is never assumed to mean anything.

The plate becomes a graph — pixel for pixel

Every plate pixel becomes a graph pixel; drags and faint smears exactly as they are

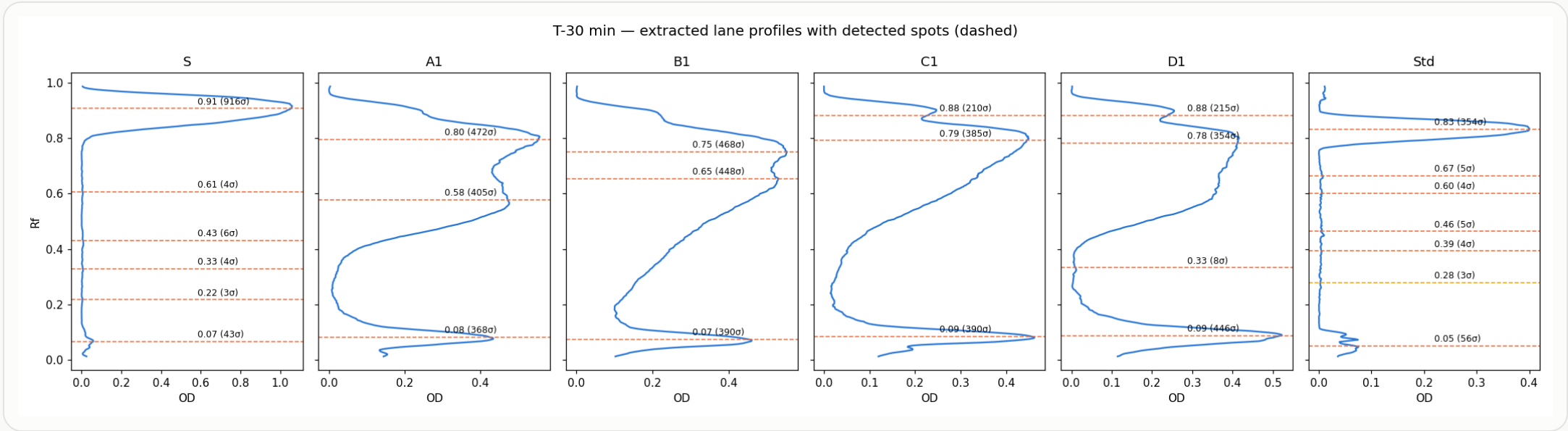


IN: normalized plates (crisp 536 px source → crisp graph; 158 px archive source → soft graph — photo quality IS graph quality). OUT: the plate on a calibrated Rf ruler. Blue line = front (Rf 1), dashed = origin (Rf 0); nothing above or below is cropped.

Three layers live here: the pixel layer (this, the trust view) · the curve layer (per-lane intensity functions — where correlations run) · the interpreted layer (classifications drawn on top).

Every spot measured — including the faint ones

Lane curves → matched-filter peaks → sub-pixel Gaussian fits, with an SNR for each

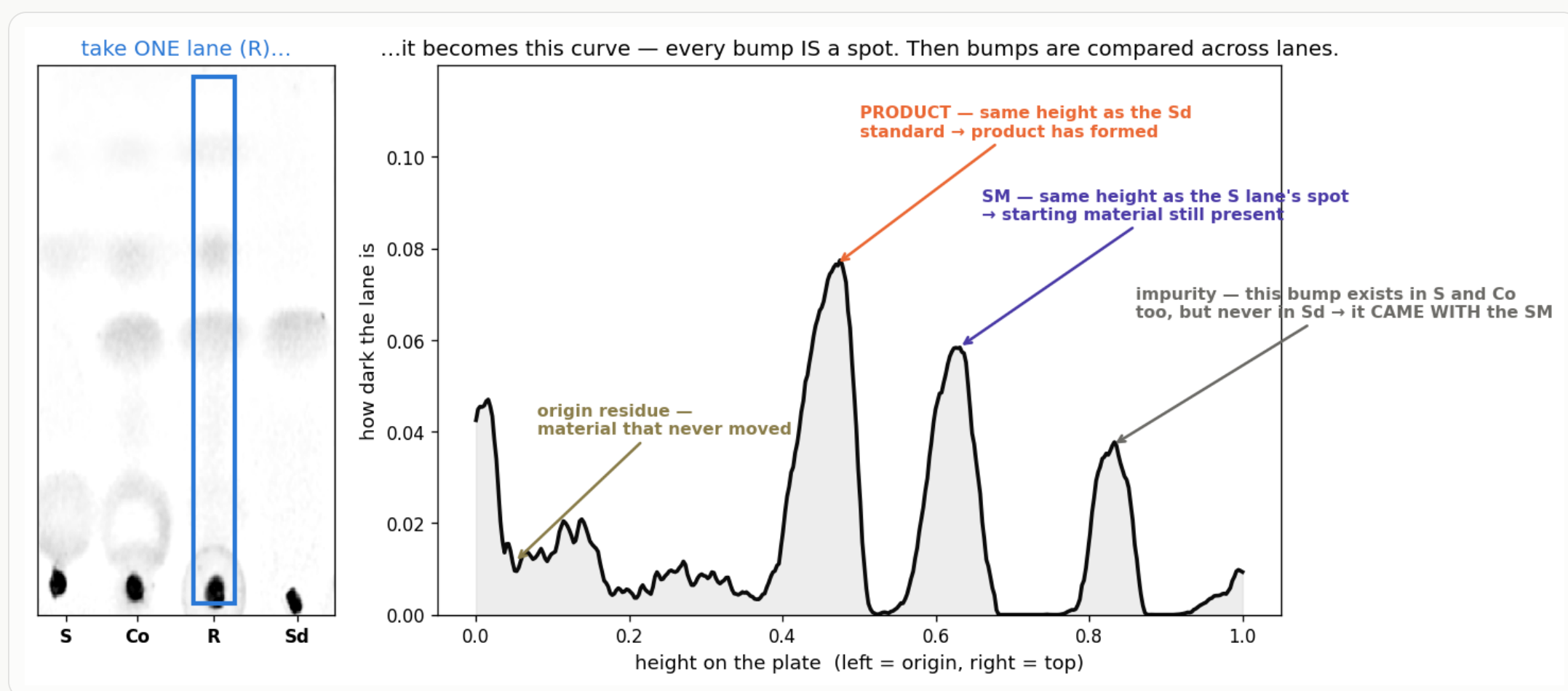


IN: the graph. OUT: every lane as a measured spot list — position (real Rf here: both lines were found), amount, width, tier. Note the S lane: four faint SM impurities at 3–6σ, and the Std lane's own impurity ladder — invisible to the eye, measured by the system.

Faint-spot machinery: absorbance (OD) signal · iterative background fit · averaging across the lane ($\sqrt{N}$ noise win) · matched filter · 2-tier confirmed/candidate reporting · cross-lane stacking rescues sub-threshold repeats.

The rules — chemist reasoning, encoded and tested

Run live on MEHQ-P33's curves; every claim below is checkable against the plate



How to read this: the blue box takes ONE lane of the plate; the right side is that same lane as a curve — every bump is a spot. The rules compare bump positions and sizes ACROSS lanes: a bump at the S lane's height = SM still there; at the standard's height = product; a bump that also appears in S and Co but never in Sd = an impurity that came with the SM.

1 · Matrix shift Δ

measured two ways (anchor + whole-curve alignment); methods must AGREE or $\Delta=0$ with widened tolerance — the guard fired on this very plate

2 · Reference anchoring

presence/absence at each reference's Δ -corrected position; anchors only from confirmed $\geq 3\sigma$ reference spots, never the origin zone

3 · Impurity inheritance

in S and Co, absent in Std → SM baggage, not a reaction by-product (action: purify SM, not the route)

4 · Co-spot as an equation

$Co \approx \alpha \cdot S + \beta \cdot R$ fitted; R^2 is the plate's self-consistency score (0.83 here)

5 · Conflict resolution

a position matching two anchors is decided by intensity consistency — chose 'product' here because $1.89 \approx Sd$'s 2.18, not the impurity's 0.07

6 · Streak guard

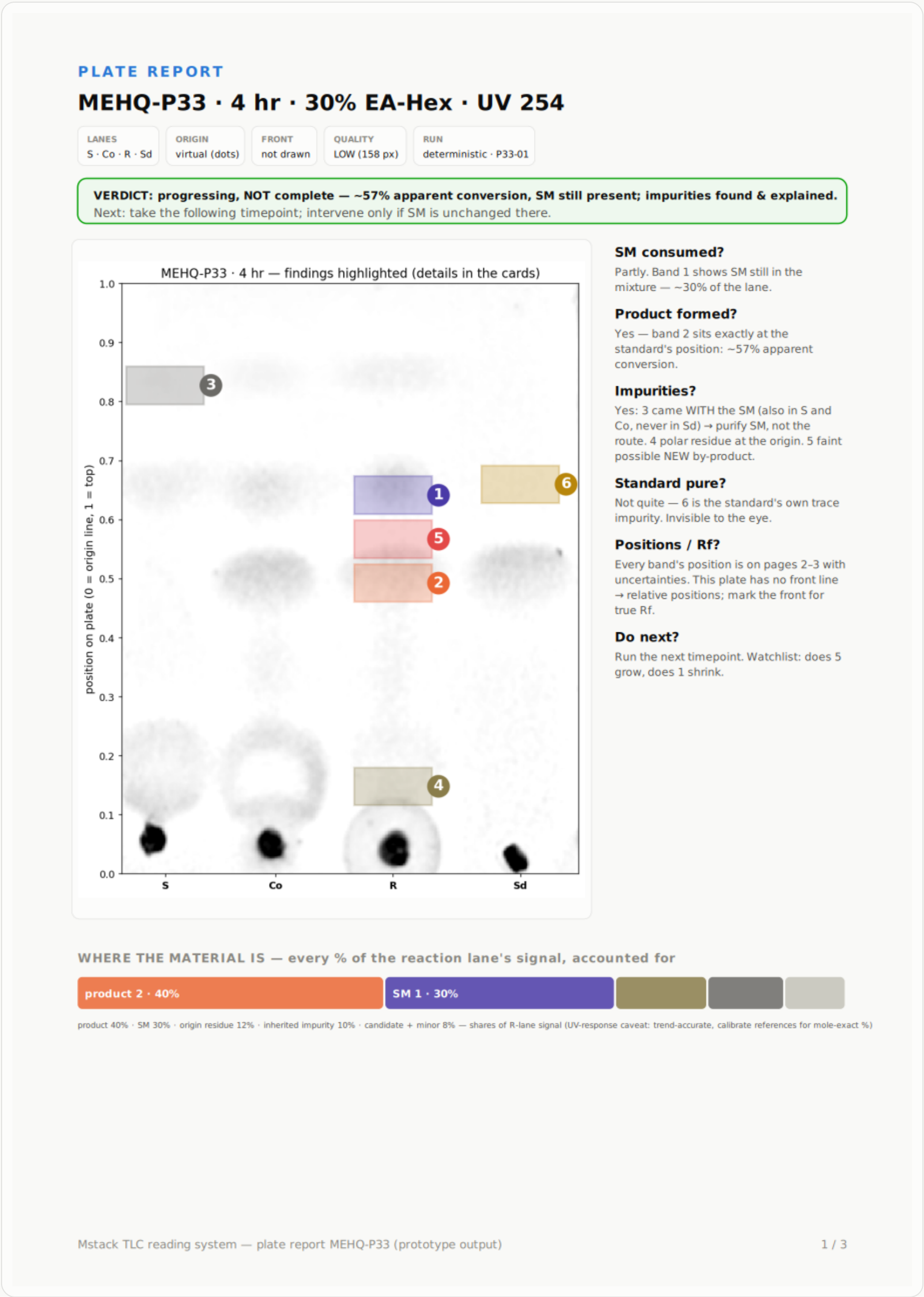
streak-dominated lanes ($>55\%$ of lane above noise) report zones only; % withheld — fired on all four T-30 condition lanes

7 · % and series verdicts

apparent conversion = product:SM within one lane (UV caveat); across timepoints $\Delta SM / \Delta product$ + zone accounting → progressing / stalled / complete / degrading

The report — answers, pointed at the plate

OUT: a 3-page report — page 1 below; pages 2-3 add finding cards, densitograms and the expert table



Confidence · determinism · trace

Why a chemist can trust it — and catch it when it is wrong

CONFIDENCE — every number has ingredients

A match's score = both spots' detection strength × position agreement × Δ quality. Verdict score = the weakest link in its evidence. High = stated. Medium = stated with its caveat. Low = the system asks instead of asserting. No front line → no Rf. No calibration → apparent % only. It never invents to avoid an empty answer: blank plates, everything-at-front, co-running references all get a named 'cannot conclude, because...'.

DETERMINISM — verified, not promised

No randomness in the pipeline; text read at temperature 0 with pinned models; outputs quantized with hysteresis so boundary values can't flip verdicts. Verified on our plates: two complete runs of T-30 produced byte-identical JSON — hash 49d796a8... both times. Every user-flagged wrong result joins a golden suite that re-runs on every change.

TRACE — the thought process is replayable

Each stage stores what it saw and decided: gate checks with values, the warped image, lines with their source, lane profiles with fits drawn on, Δ 's two measurements, and a rule-by-rule table. Disagree with a verdict → walk the chain: spot seen wrongly? (perception) · role wrong? (assignment) · wrong rule fired? (reasoning). Each error type has its own fix path.

LLM BOUNDARY — READ AND PHRASE, NEVER MEASURE OR DECIDE

Language models are used for exactly two things: reading handwriting (no classical OCR survives marker cursive — tested) and breaking a tie when a lane name is genuinely ambiguous (one cached, schema-constrained call). Never for spot detection, never for correlations, never for verdicts — those are deterministic rules, because rules can be regression-tested and trusted to repeat.

Validated on our own plates

Every stage was run against the lab's real images — the honest scorecard

Stage / rule	Result on our plates
● Plate finding + warp	6/6 plates incl. 77 px and tilted shots
● Gate rejections	person photo & chart correctly refused; both messy plates passed
● Structure (lines/lanes)	perfect on T-30 (6 lanes + both pencil lines); adaptive tuning still needed on 3 plates → user lane-map path covers them
● Faint-spot extraction	SM impurities found at 3–6σ; Std's hidden impurity ladder revealed
● Rf values	computed on T-30 with drawn lines; sub-pixel spot centres
● Δ from co-spot	disagreement guard fired correctly on P33 (weak co-spot signal at 158 px)
● Co decomposition	Co = 0.35·S + 0.99·R, R² 0.83 — plate self-consistent
● Classification + conflict rule	product vs impurity conflict resolved by intensity, correctly
● Streak guard	fired on all four T-30 condition lanes; prevented a false '100% conversion'
● Series verdict (P32 4 hr → 7 hr)	product-zone +10 pts → 'progressing', low confidence (93 px sources)
● Determinism	two full runs, hash-identical

Green = working today in the prototype. Amber = works with the documented fallback (user lane map / low-confidence trend) and is the next engineering target. Nothing on this page is projected — it all ran.